

Differential: M1 supplement

PRV 201 Frontiers of Technology (AI) · Track A: Build · Stevens Institute of Technology · September 2026

This supplement accompanies the M1 problem statement. The statement stands on its own; nothing here is needed to read it.

Industry watch, September 2026: breakthroughs, launches, regulation and investment

- **Frontier models score highly on textbook analog questions.** The top Razavi-Bench model scored 95.50%, graded by another language model, and the benchmark has an experimental simulator-assisted mode [1]. A bench tool must add what textbooks do not test: tolerances, ambiguity between faults, the cost of each measurement and safety. T5 compares the full system with such a model working alone.
- **IBM Maximo 9.2 (June 2026)** adds agentic workflows and an MCP server through which customers can connect their own agents [2], a possible route for a diagnosis engine into industrial repair depots.
- **EU AI Act transparency rules apply from 2 August 2026** [3]; the 2026 amendments moved the Annex III high-risk rules to 2 December 2027 [4] and keep a duty on providers and deployers to support their staff's AI literacy [5]. They would bind Differential only once offered in the EU.
- **Colorado re-enacted its AI law** as a regime for automated decision-making technology in consequential decisions, approved on 14 May 2026 and in force from 1 January 2027 [6]. A shop that used repair tickets to decide about its staff could come within such rules; Differential's license forbids that use.
- **Investment is flowing to AI for the trades, starting with dispatch.** Probook announced \$40 million on 23 June 2026, a \$34 million Series A led by Andreessen Horowitz and a \$6 million seed round led by Sequoia, for an AI operating system for HVAC, plumbing and electrical contractors that it built dispatch-first [7]. Our reading: investors are funding AI for scheduling and dispatch in repair businesses, while the diagnosis a technician does at the bench is not what this money buys; that is the gap Differential addresses, and repair businesses may meet AI first in their front office.

Course coverage map

Table 1. Course topics and where the project uses them, with parts named by title.

Wk	Topic	Where the project uses it	Depth
2	Introduction to AI/ML algorithms	Bayesian inference, Gaussian mixtures, boosted trees, information-gain measurement choice (Part 4 – Preliminary AI approach, risks and ethics)	Core
3	Python, the AI/ML lingua franca	The simulation, engine and evaluation code that produces every number in the statement (repository); no code appears in the statement itself	Core (code)
4	Introduction to large language models	Chat-model benchmark scores (Part 1 – Problem, stakeholders and industry context); the language-model baselines (Part 3 – Objectives and measurable success criteria; T5); why language models guess (Part 4 – Preliminary AI approach, risks and ethics)	Core
5	Building LLM-enhanced agents	An assistant whose tools act only on the diagnosis session; complaint reading and explanation (Part 4 – Preliminary AI approach, risks and ethics)	Supporting
6	Building ML systems	Pilots, locked targets with confidence bounds, stress sets, hardware protocol (Part 3 – Objectives and measurable success criteria)	Core
7	Introduction to natural language processing	Complaint reading with noise; sealed paraphrase bank (Part 3 – Objectives and measurable success criteria; T18)	Supporting
8	Image recognition systems	Meter-photo reading scored on value, unit and mode (T19)	Supporting
9	Facial recognition systems (accuracy, bias, deployment)	Error breakdown, as in face-recognition audits, by who bears the errors: complaint writers, technicians, owners of undocumented gear (Part 4 – Preliminary AI approach, risks and ethics)	Supporting
10	Autonomous transportation	Five levels of diagnostic autonomy, L0 (the technician does everything) to L4 (an unattended fixture), defined by analogy to driving automation; Differential placed at L2 on safety and accountability grounds (Part 4 – Preliminary AI approach, risks and ethics)	Supporting
11	Introduction to robotics	A sense-decide loop without actuation: instruments sense (SCPI on the 24 V boards), the engine decides the next measurement and a person acts; automated test fixtures (levels L3-L4) would add the actuation (Part 4 – Preliminary AI approach, risks and ethics)	Partial
12	AI in cybersecurity and fraud detection	Anomaly detection: the calibrated “no single fault fits” outcome flags units outside the catalog (Part 3 – Objectives and measurable success criteria; T13, T22); prompt injection, untrusted data, session-only tools and adversarial prompts (Part 4 – Preliminary AI approach, risks and ethics; T16)	Core
13	Ethics of AI: risks, responsibility and governance	Hazard audit, OSHA, liability, EU AI Act and product liability, research ethics, NIST AI RMF (Part 1 – Problem, stakeholders and industry context, Part 4 – Preliminary AI approach, risks and ethics)	Core

References

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- [3] European Commission, “Commission starts enforcing AI Act rules and new transparency requirements on 2 August (IP/26/1714),” Press release, Jul. 31, 2026. [Online]. Available: <https://digital-strategy.ec.europa.eu/en/news/commission-starts-enforcing-ai-act-rules-and-new-transparency-requirements-2-august> (accessed Sep. 28, 2026).
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- [6] Colorado General Assembly, “SB26-189, Automated Decision-Making Technology,” May 14, 2026. [Online]. Available: <https://leg.colorado.gov/bills/sb26-189> (accessed Sep. 28, 2026).
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